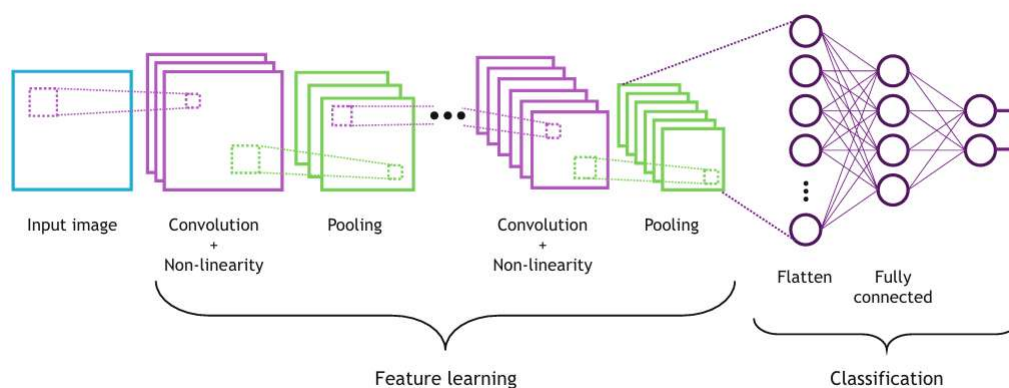


Convolutional Neural Networks (CNNs)

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Convolutional neural networks (CNNs)



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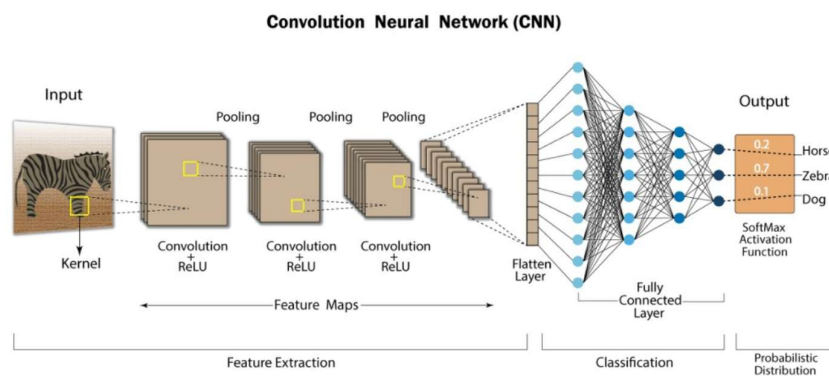
Convolutional neural networks (CNNs)

- Introduction to CNNs: Convolutional layers, Pooling layers, Filters
- Applications of CNNs: Image Recognition, Object Detection

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Convolutional neural networks (CNNs)



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Convolutional neural networks

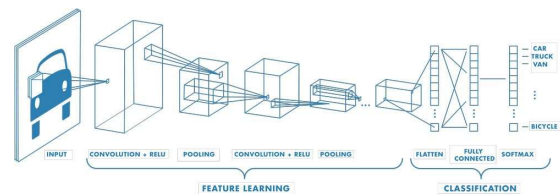
CNN's excel at identifying visual patterns and have been widely used in computer vision tasks such as object recognition, face detection, and image classification.

- Inspired by the organization of the animal visual cortex,
- Take advantage of the local hierarchy of patterns in images, allowing them to extract spatial features efficiently.
- The main difference between CNNs and traditional neural networks is the use of **convolutions**

What is a convolution?

a mathematical operation that allows the automatic extraction of important features from the images, preserving the spatial relationship between the pixels.

While a common neural network treats each pixel of an image independently, CNNs consider the neighborhood and interactions between pixels, which is crucial for visual pattern recognition.



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Convolutional neural networks - Components

CNNs are powerful deep learning models designed for processing image and spatial data. CNN's use specialized layers to detect patterns, edges, and objects, making them perfect for image recognition, and object.

Key Components of CNNs

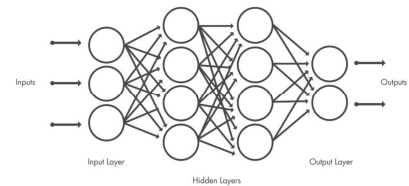
- **Convolutional Layer** – Extracts features using filters (kernels) to detect edges, textures, etc.
- **Pooling Layer** – Reduces dimensionality while keeping important features (Max Pooling is common).
- **Fully Connected Layer** – Flattens extracted features and makes predictions.
- **Activation Functions** – Uses ReLU (Rectified Linear Unit) to introduce non-linearity.
- **Softmax/Sigmoid** – Helps classify images into categories

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Convolutional neural networks (CNNs)

- A CNN is composed of an input layer, an output layer, and many hidden layers in between.
- A CNN with circles connected by many arrows, labeled from left to right: Inputs, input layer, hidden layers, output layer, and outputs.
- These layers perform operations that adjust the data with the intent of learning features specific to the data. Three of the most common layers are convolution, activation or ReLU, and pooling:
 - **Convolution** puts the input images through a set of convolutional filters, each of which activates certain features from the images.
 - **Rectified linear unit (ReLU)** allows for faster and more effective training by mapping negative values to zero and maintaining positive values. This is sometimes referred to as activation, because only the activated features are carried forward into the next layer.
 - **Pooling** simplifies the output by performing nonlinear downsampling, reducing the number of parameters that the network needs to learn.



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Convolutional neural networks (CNNs)

CNN's

- Automatically learn spatial hierarchies of features
- Reduce computational cost with parameter sharing
- Work exceptionally well in computer vision tasks
- are often used for classification and computer vision tasks.
- before CNNs, manual and time-consuming methods of resource extraction were used to identify objects in images.
- a more scalable approach to image classification and object recognition tasks, leveraging the principles of linear algebra, to identify patterns in an image.
- can be compute-intensive, which requires graphics processing units (GPUs) to train models.

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Convolutional neural networks (CNNs)

Convolutional neural networks (CNNs) are a type of neural network that is specifically designed for image and video processing tasks. They are inspired by the visual cortex of the human brain, which is composed of multiple layers of neurons that are sensitive to different spatial and temporal features in the visual input.

CNNs consist of multiple layers of convolutional and pooling layers.

The convolutional layers are used

- to extract features from the input image, while the pooling layers are used
- to reduce the dimensionality of the feature maps
- to increase the robustness of the features to small translations and distortions in the input image.

The main building block of a CNN is the **convolutional layer**, which applies a set of learnable filters to the input image.

- These filters are used to extract different types of features from the image, such as edges, textures, and shapes.
- The output of the convolutional layer is a feature map, which is a multi-dimensional array of numbers that represents the presence of the learned features in the input image.

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Convolutional neural networks

CNNs are trained using supervised learning algorithms, such as backpropagation, to learn the relationship between the input image and the output labels.

The training process involves adjusting the weights and biases of the filters in the convolutional layers, so that the output of the network matches the desired output.

CNNs have been successfully used in a wide range of computer vision tasks such as image classification, object detection, semantic segmentation, and object tracking.

They can also be used for tasks such as natural language processing, speech recognition, and generative models.

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CNN's: Key Characteristics

Local receptive fields: CNNs employ small, machine-learning filters that move across the input image, focusing on specific regions to detect local features such as edges, textures, and patterns.

Weight sharing: The same filter is applied to various regions of the input image, decreasing the number of parameters and computational complexity while allowing the network to recognize objects regardless of position.

Pooling: Pooling layers minimize the spatial dimensions of feature maps, making the network more efficient and adaptable to changes. Max pooling, for example, takes the highest value from each patch, aggregating feature presence while decreasing parameters.

CNNs' structure, which combines local receptive fields, weight sharing, and pooling, makes them extremely efficient and reliable for image processing tasks.

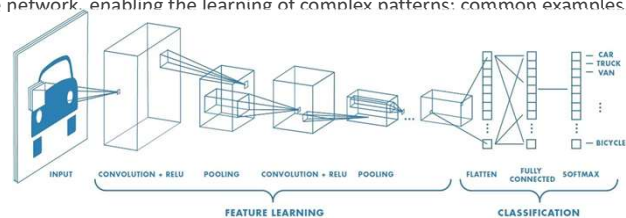
This enables CNNs to perform tasks such as image classification, object detection, and segmentation, making them essential in computer vision.

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CNN's: Layers

- **Convolutional Layers:** Apply filters to the input image to extract local features such as edges and textures.
- **Pooling Layers:** Reduce the spatial dimensions of feature maps, decreasing the computational load and enhancing feature robustness.
- **Fully Connected layers:** Connect every neuron in one layer to every neuron in the next, integrating extracted features for final predictions.
- **Dropout:** Randomly sets a fraction of neurons to zero during training to prevent overfitting and improve model generalization.
- **Activation Functions:** Introduce non-linearity into the network, enabling the learning of complex patterns: common examples include ReLU and Sigmoid.

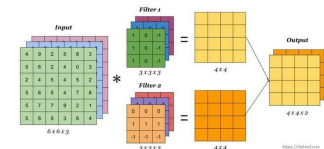


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Convolutional Layers

- The basis of CNNs, designed to automatically and adaptively learn spatial hierarchies of features from input images.
- These layers perform a mathematical operation called **convolution**, which involves sliding a **filter** or **kernel** over the input image to produce a feature map.
- This process allows the network to focus on small, distinct features in the initial layers and progressively learn more abstract and comprehensive features in deeper layers.
- The primary role of convolutional layers is **feature detection**. By applying various filters (e.g., for edge detection, texture analysis, etc.),
 - these layers can identify various features within an image, such as edges, corners, or textures, which are crucial for understanding the image's content.
 - particularly effective for tasks that involve image recognition, classification, and analysis.



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Activation functions

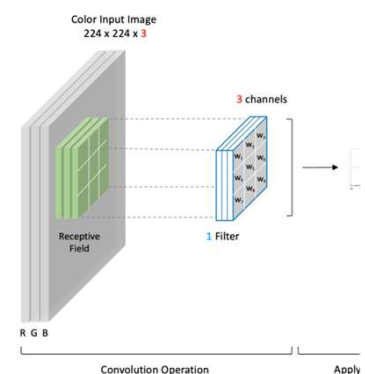
Activation functions introduce non-linear properties to the network:

- enabling it to learn complex patterns and perform tasks beyond mere linear classification.
- help decide which neurons should be activated, thereby determining the output of the network for a given set of inputs.

Activation functions allow neural networks to capture nonlinear relationships in the data, making them powerful tools for a wide range of applications.

The most commonly used activation functions in CNNs are:

- **ReLU** (Rectified Linear Unit): The ReLU function is defined as $f(x) = \max(0, x)$. It has become the default activation function for many types of neural networks because it introduces non-linearity while being computationally efficient. ReLU is particularly popular in CNNs due to its ability to mitigate the vanishing gradient problem, allowing models to learn faster and perform better.
- **Sigmoid**: The Sigmoid function, defined as $f(x) = 1/(1+e^{-x})$, outputs a value between 0 and 1. This characteristic makes it suitable for binary classification tasks or as the final activation function in a network predicting probabilities. However, its use in hidden layers has declined due to the vanishing gradient problem, especially in deep networks.



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Pooling layers

Pooling layers are integral to the architecture of CNNs, serving a pivotal role in reducing the spatial size of the representation, decreasing the amount of parameters and computation in the network.

This is known as downsampling or subsampling, helps in making the detection of features invariant to scale and orientation changes, ultimately enhancing the network's efficiency and performance.

- **Max Pooling:** This is the most commonly used form of pooling in CNNs, where the maximum value from each cluster of neurons in the feature map is taken. Max pooling helps in highlighting the most prominent features, effectively reducing noise and computational load for the network.
- **Average Pooling:** Contrary to max pooling, average pooling calculates the average value of each cluster, smoothing out the feature map. While less common than max pooling, it is useful in certain contexts where preserving the background information is important.

Pooling layers, by reducing the dimensions of the feature maps,

- help in reducing the computation but also contribute to the network's ability to generalize by providing an abstracted form of the features.
- reduces the sensitivity of the output to small variations and distortions in the input image.

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Dropout Layers

Overfitting is a common pitfall where a model learns the training data too well, including its noise and outliers, resulting in poor performance on unseen data. The overfitting indicates that the model has lost its generalization ability, making it crucial to implement strategies to prevent overfitting for robust model performance. One such effective strategy is **the use of dropout layers for regularization**.

Dropout layers work by randomly "dropping out" a set percentage of neurons in the network during training at each update cycle, preventing them from participating in forward and backward propagation.

This technique forces the network to not rely on any single neuron and spreads out importance among many, thus reducing overfitting.

Dropout can be seen as a method of training a large ensemble of neural networks with shared weights, where each iteration uses a slightly different network configuration.

- dropout rate, which correspond to the % of neurons to drop, is a critical decision.
- A rate of 0.5 is a good starting point for hidden layers, ...
- this value should be adjusted based on the specific problem and dataset.
- For input layers, a lower rate such as 0.2 is often used.
- important to experiment with different rates to find the optimal configuration for your model.

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Fully Connected Layers

- After the **convolutional** and **pooling layers** have done their job of feature extraction and dimensionality reduction, the role of making sense of these extracted features falls to the fully connected (dense) layers.
- **Fully connected layers** play a crucial role, especially in classification tasks, where the goal is to classify the input image into various categories based on the learned features.
 - Each neuron in these layers is connected to every neuron in the previous layer,
 - they work by combining these features in various ways to make predictions.
 - while convolutional layers are good at identifying features within input data, **fully connected layers** are responsible for the high-level reasoning required to make final predictions.
 - For example, in an image recognition task, while convolutional layers might identify edges, textures, and colors, it's the fully connected layers that decide whether the combination of these features constitutes a particular object.

Before the extracted features can be passed into the fully connected layers, they must be flattened since fully connected layers expect a one-dimensional vector of data, whereas the output from the convolutional and pooling layers is a multi-dimensional tensor.

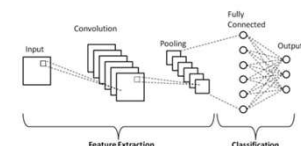
Flattening is simply the process of converting this multi-dimensional tensor into a one-dimensional vector.

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Convolutional neural networks

- Convolutional neural networks are distinguished by their high performance with image, speech, or audio signal inputs. They have three main types of layers:
 - **Convolutional layer**
 - Filters
 - Stride
 - Padding
 - Activation Function
 - **Grouping Layer**
 - **Fully Connected Layer (FC)**
- The convolutional layer is the first layer of a convolutional network. While convolutional layers can be followed by additional convolutional layers or clustering layers, the fully connected layer is the final layer.
- With each layer, CNN increases its complexity, identifying larger parts of the image. Previous layers focus on simple elements such as colors and borders. As the image data progresses through the CNN layers, it begins to recognize larger elements or shapes of the object until it finally identifies the intended object.



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Convolutional layer

- The convolutional layer is the primary component of a CNN.
- The core operation of convolutional layers is the convolution operation. It involves applying a filter (also known as a **kernel** or **convolutional kernel**) to small regions of the input data.
- Each filter extracts different features from the input data, such as edges, textures, or higher-level patterns.
- Convolutional layers may also include activation functions (ReLU) to introduce non-linearities and increase the model's representational capacity.
- Multiple convolutional layers may be stacked to learn increasingly complex features.

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Convolutional neural networks

- A convolutional neural network can have tens or hundreds of layers that each learn to detect different features of an image.
- Filters are applied to each training image at different resolutions, and the output of each convolved image is used as the input to the next layer.
- The filters can start as very simple features, such as brightness and edges, and increase in complexity to features that uniquely define the object.

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Convolutional neural networks - Architectures

- LeNet
- AlexNet
- VGGNet
- GoogLeNet
- ResNet
- ZFNet

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Advanced CNN Architectures

- **ResNet (Residual Neural Network)**: Explore the concept of residual learning, where the network learns residual functions to optimize the learning process.
- **InceptionNet (GoogLeNet)**: Unravel the inception module, which incorporates multiple convolutional filters of different sizes within a single layer.
- **XceptionNet**: Dive into this architecture that combines the depthwise separable convolution and the inception module for better performance.
- **MobileNet**: Understand the lightweight design of MobileNet, optimized for mobile and embedded devices.

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Advanced Training Techniques

To improve the performance and generalization of CNNs, advanced training techniques are employed:

- **Transfer Learning:** Learn how to leverage pre-trained CNN models, fine-tuning them for specific tasks, saving training time and resources.
- **Data Augmentation:** Discover how data augmentation can increase the diversity of the training data, reducing overfitting and improving the model's robustness.

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Convolutional neural networks train

To train convolutional neural networks for image analysis it is possible to use the following:

- Training the model from scratch;
- Using transfer learning (based on the idea that you can use knowledge of one type of problem to solve a similar problem);
- Using a pretrained CNN to extract features for training a machine learning model.

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Applications of convolutional neural networks

- Facial recognition software, image classification, speech recognition programs, are the most familiar, and comprise a major part of our everyday life, especially with image-savvy social media networks. Some of the key applications of CNN:
 - Decoding Facial Recognition
 - Analyzing Documents
 - Collecting Historic and Environmental Elements
 - Understanding Climate
 - Understanding Gray Areas
 - Advertising
 - Other Interesting Fields

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CNN applications

CNNs provide an optimal architecture for uncovering and learning key features in image and time-series data. Main applications such as:

- **Medical Imaging:** CNNs can examine thousands of pathology reports to visually detect the presence or absence of cancer cells in images.
- **Audio Processing:** Keyword detection can be used in any device with a microphone to detect when a certain word or phrase is spoken ("Hey Siri!"). CNNs can accurately learn and detect the keyword while ignoring all other phrases regardless of the environment.
- **Object Detection:** Automated driving relies on CNNs to accurately detect the presence of a sign or other object and make decisions based on the output.
- **Synthetic Data Generation:** Using Generative Adversarial Networks (GANs), new images can be produced for use in deep learning applications including face recognition and automated driving.

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